

Digital Lending & Real-Time Credit Decisioning

Created WhiteQuest, an end-to-end digital lending platform designed to automate onboarding, KYC, credit assessment, risk-based decisioning and lender-side monitoring.

Real-time	decision engine
4	decision outcomes
5	test profiles
100%	decision automation

THE CORE IDEA

Replace fragmented, manual underwriting with a transparent rule-driven lending journey — while giving higher-risk borrowers controlled alternatives instead of blanket rejection.

WHAT WAS BUILT

Component	What the platform does
Borrower onboarding	Structured digital application and multi-step application flow
External data layer	KYC, PAN, bureau, bank-statement and eSign integrations designed around financial infrastructure
Business Rule Engine	Real-time borrower risk evaluation and automated decisioning
Risk-based pricing	Reduced-principal or risk-premium alternatives for near-prime borrowers
Admin Command Center	Portfolio monitoring, application drill-down and decision audit trail

Project lens: combines product design, financial mathematics, credit policy, automation and backend state management into one lending workflow.

01 • PLATFORM ARCHITECTURE

A complete lending journey, designed around automated decisions

The platform was structured to mirror a production-style digital lending ecosystem, while the current demo uses simulated API calls for external services.

THE LENDING FLOW

Stage	System behaviour	Decision / output
01 • Onboarding	Borrower enters personal, employment and loan-request information through a structured multi-step flow.	Application created
02 • KYC & identity	Aadhaar eKYC / DigiLocker and PAN verification are designed to validate identity fields.	Verified / KYC_FLAG
03 • Financial data	Bank-statement / Account Aggregator layer provides verified income and existing obligations.	Income + EMI inputs
04 • Credit bureau	CIBIL / Experian layer supplies credit score and historical risk information.	Credit score
05 • Credit engine	Business Rule Engine evaluates score, DTI and identity conditions sequentially.	Approve / conditions / reject
06 • Pricing	Near-prime cases receive a principal-reduction or risk-premium alternative.	Offer options
07 • Monitoring	Admin Command Center receives application milestones and decision details.	Audit + portfolio view

EXTERNAL DATA LAYER

Integration	Designed purpose
Aadhaar eKYC	UIDAI / DigiLocker — extract and verify name, DOB, gender and address
PAN verification	NSDL / Income Tax Department — identity validation and bureau retrieval
Credit bureau	TransUnion CIBIL / Experian — score and historical default information
Bank statement	Account Aggregator framework through vendors such as Perfios / Finbox
eSign & agreement	Aadhaar eSign / NSDL e-Gov for digitally binding agreements

Financial mathematics meets transparent credit policy

Every approval is grounded in explicit calculations and sequential rules, making the decision path auditable rather than dependent on opaque manual underwriting.

CORE BUSINESS MATH

Metric	Formula	Purpose
EMI	$EMI = [P \times R \times (1+R)^N] / [(1+R)^N - 1]$	Calculates the monthly repayment for the proposed loan
Total Interest	Total Interest = (EMI × N) – P	Shows the borrower's total cost of borrowing
DTI / FOIR	$DTI \% = ((Existing\ EMIs + New\ Proposed\ EMI) / Verified\ Monthly\ Income) \times 100$	Measures affordability and debt burden

THE BUSINESS RULE ENGINE

Outcome	Trigger logic	System action
KYC_FLAG	Aadhaar name does not match PAN name	Stop auto-approval and route to Admin Portal for manual verification
REJECTED	Credit Score < 600 OR DTI ≥ 50%	Instant hard rejection to protect portfolio quality
APPROVED_CONDITIONS	Credit Score 650–749 OR DTI 40–49%	Offer controlled alternatives through risk-based pricing
APPROVED	Credit Score ≥ 750 AND DTI < 40%	Instant approval for full amount at base rate

RISK-BASED PRICING

DTI Optimisation: reduce principal by 20% while keeping the base rate.
Risk Premium: grant the full requested amount with a +6% interest premium to compensate for elevated risk.

03 • TEST PROFILES

Five borrower journeys, five different decision paths

The platform was tested against profiles designed to trigger prime, near-prime, sub-prime, returning-borrower and concurrent top-up scenarios.

CASE STUDIES

Borrower	Profile / request	Engine result
Robert	₹1,20,000 income • ₹5,000 existing EMI • CIBIL 782 • ₹1,50,000 / 12 months	New EMI ₹13,327 • Total EMI ₹18,327 • DTI 15.2% • APPROVED
Chris	₹45,000 income • ₹8,000 existing EMI • CIBIL 715 • ₹2,50,000 / 36 months	New EMI ₹9,041 • Total EMI ₹17,041 • DTI 37.8% • APPROVED_CONDITIONS
Benedict	CIBIL 580 • 3 previous defaults	Hard rejection triggered by score <600 • REJECTED
Mark — Cleared Loan	Returning user • previous 6-month loan fully paid • CIBIL 760 • ₹1,60,000 income	Historical paid-off loan excluded from active DTI • identity verified • APPROVED
Mark — Active Top-Up	Active ₹3,00,000 loan • ongoing EMI ₹33,685 • requests ₹1,50,000 top-up	New EMI ₹13,327 • Total EMI ₹47,012 • DTI 29.38% • APPROVED

WHAT THESE CASES DEMONSTRATE

Scenario	Product behaviour
Prime borrower	Frictionless full approval when both score and affordability are strong
Near-prime borrower	Controlled alternatives instead of a blanket rejection
Sub-prime borrower	Hard risk floor prevents unsafe lending
Returning borrower	Verified identity supports a faster repeat journey
Active top-up	System carries concurrent obligations into affordability calculations

From borrower journey to lender-side command center

WhiteQuest does not stop at the approval screen. The architecture includes state management, milestone syncing and a lender-facing view for portfolio oversight.

SYSTEM CONTROL LAYER

Capability	Implementation
Frontend state machine	React Context API manages the complex, multi-step application state
Real-time cross-tab syncing	Browser local-storage events sync borrower milestones to the Admin Portal
Precision tracking	Custom tracking prevents ghost applications when users only view their dashboard
Admin Command Center	Credit officers can view approval ratios, total disbursed and individual application drill-downs
Decision audit	Admin can audit DTI calculations, CIBIL score and bank-statement inputs

PRODUCT DESIGN PRINCIPLES

01	Transparent Every decision is tied to explicit rules and financial calculations.
02	Risk-aware Near-prime borrowers receive controlled alternatives instead of automatic rejection.
03	Automation-first The Business Rule Engine removes repetitive manual underwriting from the core journey.
04	Operationally visible The Admin Command Center makes applications and decisions traceable.

THE BUILD IN ONE LINE

Created WhiteQuest as an end-to-end digital lending platform combining KYC, credit assessment, affordability mathematics, automated decisioning, risk-based pricing, returning-borrower logic and lender-side monitoring.